

One-Class Classification Constraint in Reconstruction Networks for Multivariate Time Series Anomaly Detection

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Abstract—Detecting the anomalies in multivariate time series (MTS) data is crucial for maintaining the stability of industrial manufacturing processes and biochemical operations. However, current methods often focus on capturing the normal patterns of training data while overlooking the potential of latent representations. This research introduces the hypersphere constraint network (HSC), an innovative self-supervised model for anomaly detection in MTS. This approach uniquely integrates a one-class classification framework to regulate latent distribution. First, the HSC employs a temporal convolutional network (TCN) and a multilayer perceptron (MLP) to extract latent representations of input data, imposing constraints on the latent distribution to achieve a one-class loss. Second, a self-attention mechanism is applied to reconstruct the input data and calculate the reconstruction loss. Anomalies are identified by combining the one-class loss with the reconstruction loss. By integrating one-class classification with a reconstruction-based method, the HSC significantly increases sensitivity to anomalous data, enhancing the distinction between normal and abnormal data. Evaluations on three real-world datasets and a simulated dataset demonstrate that the HSC model outperforms existing state-of-the-art methods in anomaly detection.

Index Terms—Anomaly detection, deep learning, multivariate time series (MTS), one-class classification, transformer.

NOMENCLATURE

$\mathbf{X}_{\text{input}}$	Input multivariate time series data.
$\mathbf{x}_{\text{int}}$	Input data at time t .
x_t^n	Input data at time t for the n th variable.
$\mathbf{y}$	Output label vector representing anomalies.
y_t	Output label at time step t .
$\mathbf{X}$	Input matrix for the encoder.
$\mathbf{x}_n$	Feature vector for the n th variable.
$\mathbf{c}$	Center of the hypersphere in latent space.
$\mathbf{Z}$	Latent representation matrix.
$\mathbf{z}$	Latent representation.

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α	Attention score in the self-attention mechanism.
α_{att}	Linear feedforward network. Used to calculate the attention score.
$\sigma(\cdot)$	Activation function (e.g., ReLU).
$f_{\text{encoder}}(\cdot)$	Encoder function.
H	Number of heads used in multihead attention.
R	Radius of the hypersphere in latent space.
$\hat{\mathbf{X}}$	Reconstructed input data.
$\hat{\mathbf{x}}_t^n$	Reconstructed feature data at time step t for the n th variable.
$\mathbf{W}_{\text{MLP}}, \mathbf{b}_{\text{MLP}}$	Learnable parameters in MLP.
β	Hyperparameter balancing one-class and reconstruction loss.

I. INTRODUCTION

MULTIVARIATE time series (MTS) data are of great significance across various domains. In modern industrial production, data continuously monitored by sensors reflect the operating conditions of manufacturing processes [1]. Similarly, in aerospace missions, hundreds of sensors track the real-time performance of an aircraft [2], [3]. Such sophisticated systems require efficient, reliable, and continuous operations, as even subtle anomalies can lead to severe system failures and significant economic loss [4]. Therefore, prompt identification and early warning of potential anomalies in MTS are critical. Efficient MTS anomaly detection methods can assist in timely discovery of system abnormalities, thereby prompt identification and early warning of potential anomalies in MTS are critical. Efficient MTS anomaly detection methods averting drastic consequences.

Anomaly detection in MTS refers to the identification of data that deviate from the expected distribution within a collection of interconnected time series [5]. The main challenges in MTS anomaly detection are threefold. First, in real industrial environments, most data are in a normal state, making it labor-intensive to label anomalous data, which render supervised methods unsuitable [6]. Second, MTS anomaly detection involves capturing both complex relationships among variables and temporal dependencies. Therefore, an effective MTS anomaly detection method must capture both spatial and temporal relationships [7]. Lastly, the model must exhibit high sensitivity to the distribution of anomalous data and give a

clear distinction between outputs for normal and anomalous data [8], [9].

To address the MTS anomaly detection problem, several feature-based methods from classical unsupervised machine learning have been developed. OC-SVM [10], [11] reframes the anomaly detection task as a one-class classification challenge, while SVDD [12] projects latent representations of data into a hypersphere, classifying normal and anomalous data based on their positions within this hypersphere. Clustering strategies such as K -means [13] and DBSCAN [14] designate samples as anomalies when their distance from cluster centroids exceed a predefined threshold. Principal component analysis (PCA) can be employed to extract statistical models of normal data, with anomalies identified based on deviations from principal component scores [15]. TTAD [16] uses K -means [17] or SMOTE [18] algorithms to augment the test data, thereby boosting the efficiency and robustness of anomaly detection. These methods use an unsupervised approach to achieve anomaly detection. However, they only capture relationships among variables and neglect temporal relationships.

Recently, deep learning has gained widespread attention for its exceptional ability to capture both intervariable relationships and temporal dependencies simultaneously. Current mainstream MTS anomaly detection methods can be divided into three categories: prediction-based methods, reconstruction-based methods, and distribution-based methods [19], [20].

A. Prediction-Based Methods

Detect anomalies by analyzing prediction errors. LSTM-NDT [21] employs LSTM [22] networks and combines them with unsupervised dynamic thresholding techniques for anomaly detection. GDN [23] utilizes graph neural networks to assess the interconnections across sensors and performs time series prediction based on attention mechanisms. Deviations between predicted and actual results are then used to detect anomalies. The GTA [24] method leverages a Transformer-based architecture to automatically learn graph structures, apply graph convolution, and model temporal dependencies for effective MTS anomaly detection. DTAAD [25] model combines temporal convolutional network (TCN) and Transformer models [26] and improves prediction accuracy through global and local temporal convolutions via Dual TCN. Additionally, it leverages meta-learning to enhance anomaly detection performance in scenarios with small sample data.

B. Reconstruction-Based Methods

Identify anomalies by analyzing reconstruction errors [27]. DAGMM [28] combines a deep autoencoder and a Gaussian mixture model to identify anomalies in MTS by modeling data distributions and detecting deviations using energy-based representations. MEGA [29] leverages multi-scale discrete wavelet transform and graph convolutional network modules to enhance autoencoder architecture, effectively capturing subtle anomalies. KalmanAE [30] leverages an autoencoder-optimized Kalman filter to adaptively estimate

normal system states, using deep embeddings to optimize filter parameters and detect anomalies based on reconstruction errors from the filtered signal. USAD [5] employs a dual reconstruction strategy with variational autoencoders (VAEs) to learn the normal data distribution and detect anomalies by identifying deviations from these learned distributions. In MAD-GAN [31], LSTM functions as both generator and discriminator in a GAN [32], combining the discriminator's judgment with the generator's reconstruction loss to score anomalies. The TranAD [33] method employs an encoder-decoder structure with adversarial training to amplify reconstruction errors and improve detection performance. CAE-AD [34] utilizes a contrastive autoencoder framework with data augmentation in both time and frequency domains to achieve robust representations. MANomaly [35] employs a mutual adversarial network for data reconstruction and implements a high anomaly suppression mechanism to enhance its effectiveness. D3R [36] uses noise diffusion and a designed attention structure to address the challenges of MTS anomaly detection in unstable environments. AT-DCAEP [37] combines convolutional autoencoders and attention mechanisms to better capture temporal relationships. In addition to the methods mentioned above, recent approaches have integrated graph neural networks to capture relationships among non-Euclidean data variables [38], [39]. For example, in the GReLeN [40], the VAE [41] serves as an overall framework, and graph neural networks are integrated to learn dependencies across different features. MTAD-GAT [42] leverages graph attention networks (GATs) to capture both temporal dependencies and intervariable relationships for effective MTS anomaly detection.

C. Distribution-Based Methods

Introduce normalizing flows, which use a series of invertible transformations to map complex distributions to simpler ones, thereby estimating data probability [43], [44]. A lower probability density indicates a higher likelihood of anomalies. GANF [45] integrates Bayesian networks with normalizing flows to model conditional dependencies among time series, enabling effective density estimation and anomaly detection in MTS. MTGFlow [46] employs dynamic graph structure learning and entity-aware normalizing flows for unsupervised anomaly detection, enhancing anomaly detection. Compared to classical methods, deep learning approaches can better capture the temporal features of data.

However, these methods primarily focus on capturing the characteristics of normal data, without emphasizing the distinctions between normal and anomalous data and may not fully utilize latent data representations.

To address these existing issues, this article proposes the hypersphere constraint network (HSC) model for anomaly detection. First, to better understand the temporal dependencies, the HSC model uses a TCN encoder to identify temporal dependencies. Second, to fully leverage the data in the hidden space, the model applies constraints within the latent representation, differentiating normal from abnormal data to amplify the reconstruction loss. Finally, to better extract the

features in the latent space, the decoder uses a self-attention mechanism that combines the GAT [47] and transformer model [26] to effectively extract features and reconstruct the input data. Finally, anomaly detection is achieved by merging the reconstruction and one-class classification losses.

The contributions of this research are as follows.

- 1) This study introduces a novel network model named HSC. By integrating one-class classification with a reconstruction-based neural network, the HSC model effectively separates normal and anomalous data in the latent space. This separation amplifies reconstruction errors and enhances the model's anomaly detection capability.
- 2) To enable effective reconstruction of the input data and capture the relationships between multivariate data, HSC utilizes a self-attention mechanism that combines the GAT and Transformer models to extract features in the latent representation.
- 3) In evaluations conducted across three publicly available real-world datasets and six penicillin fermentation test datasets, the HSC model outperformed other state-of-the-art methods. These results underscore the effectiveness of the HSC model in anomaly detection tasks.

The remainder of this article is organized as follows: Section II presents the preliminary knowledge of HSC. Section III elaborates on the details of the proposed method. Section IV evaluates the effectiveness of the HSC model on three public datasets and six penicillin fermentation test datasets, assessing its ability to amplify the distinction between anomalous and normal data. Section V discusses the scalability and robustness of the HSC model and explores potential future research directions. Finally, Section VI provides a summary of this study.

II. PRELIMINARY WORK

In this study, two key methods are employed: TCN [40] and an attention mechanism. Both approaches have proven effective in time series analysis.

A. TCN Model

The TCN model, introduced by Bai et al. [48], is an effective method for capturing temporal data features using convolutional networks. The TCN consists of three key components: causal convolutions [49], dilated convolutions [50], and residual connections [51].

Causal convolution [49] prevents the leakage of future information by introducing an appropriate offset during the convolution operation. This ensures that the output y_t depends only on the current and past data. The corresponding formula is given as follows:

$$y_t = \sum_{k=0}^{K-1} w_k \cdot x_{t-k} \quad (1)$$

where x_{t-k} represents the data at time $t - k$, K denotes the kernel size, and w_k is the convolution kernel weight.

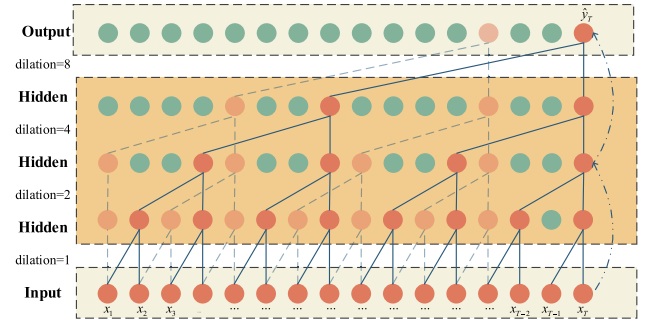


Fig. 1. Overall structure of the TCN model.

Dilated convolution [50] enables TCN to capture long-range dependencies in sequence data. The formula is expressed as follows:

$$y_t = \sum_{k=0}^{K-1} w_k \cdot x_{t-d \cdot k} \quad (2)$$

where d is the dilation factor, i is the current layer in the hidden state, and $d = 2^i$. The greater the number of hidden layers, the longer the temporal dependencies that can be captured.

Residual connections [51] are widely used in deep learning to mitigate the vanishing gradient problem. To capture longer sequences, TCN require stacking of multiple layers. To maintain training stability, TCN also incorporate residual connections. The formula is expressed as follows:

$$y = \sigma(x + \mathcal{F}(x)) \quad (3)$$

where x is the input, $\mathcal{F}(x)$ represents the transformation applied to x , and σ is the activation function. The addition $(x + \mathcal{F}(x))$ provides a direct gradient path, improving the training stability of deep networks.

The overall architecture of the TCN is illustrated in Fig. 1. The orange circles represent the nodes observed at the output y_t . Through multiple layers of dilated convolutions, the model effectively captures long-range sequence features.

B. Attention Mechanism

The design of the attention mechanism is inspired by the cognitive abilities of humans, where information processing is not uniform across all data. Humans identify key information and allocate additional cognitive resources to processing and understanding crucial aspects. Similarly, the attention mechanism extracts significant features and allocates resources efficiently [52].

The canonical attention mechanism proposed by Vaswani et al. [26] is used to overcome the limitations of traditional RNN methods and capture information from all positions in long sequence. The principle of obtaining an attention score is given by

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (4)$$

where d_k is a scaling factor. Q , K , and V represent the query, key, and value, respectively, derived from the input data to calculate attention scores.

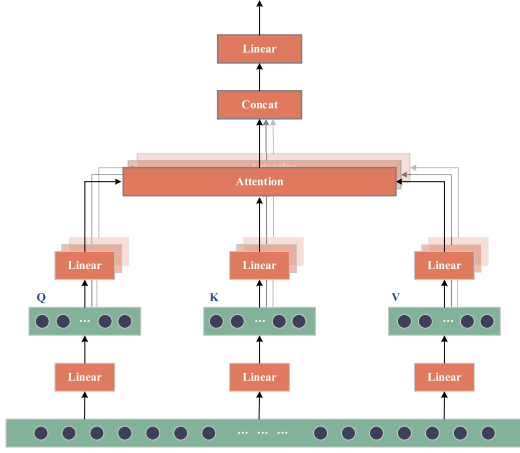


Fig. 2. Overall structure of the multihead attention mechanism.

To enhance model expressivity, the multihead attention mechanism is often used. This mechanism is a fundamental and pivotal module within the Transformer model. It runs the attention mechanism in parallel multiple times, applying different linear transformations to the original Q , K , and V . This approach captures various aspects of the input data, enriching the representation of the input. The concept can be formulated as follows:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (5)$$

where W_i^Q , W_i^K , W_i^V are the projection matrices for the Q , K , and V , respectively, for each attention head i . The operation of multihead attention is illustrated in Fig. 2.

III. PROPOSED METHOD

A. Problem Statement

In the MTS anomaly detection problem, the input data are denoted by $\mathbf{X}_{\text{input}} = (\mathbf{x}_{\text{in}1}, \mathbf{x}_{\text{in}2}, \mathbf{x}_{\text{in}3}, \dots, \mathbf{x}_{\text{in}T}) \in \mathbb{R}^{T \times N}$, where T represents the number of timestamps in the input time series and N equals the number of features. The data at each timestamp t are represented as $\mathbf{x}_{\text{int}} = (x_t^1, x_t^2, x_t^3, \dots, x_t^N) \in \mathbb{R}^N$, where $N = 1$ for univariate time series and $N \geq 1$ for MTS. The primary aim of MTS anomaly detection is to identify unusual deviations within the data.

Following the data processing method described in previous works [23], [31], [40], [42], continuous data segments $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_L) \in \mathbb{R}^{L \times W \times N}$ are used as inputs for HSC. These segments are obtained by applying a sliding window method, characterized by a window size W , a stride S , and the number of segments $L = (\text{floor}(T - W)/S) + 1$, where $\text{floor}(\cdot)$ denotes rounding down.

The output is $\mathbf{y} = (y_1, y_2, y_3, \dots, y_T) \in \mathbb{R}^T$, $y_t \in \{0, 1\}$. This vector represents the anomaly detection results at each timestamp. Specifically, an output value of 1 indicates an anomaly at time t , while a value of 0 signifies that no anomaly is detected.

The variables used in our proposed method are summarized in Nomenclature.

B. Model Overview

The HSC model utilizes an encoder–decoder architecture, as illustrated in Fig. 3, and is a reconstruction-based neural network designed for anomaly detection in MTS.

Encoder: The encoder processes the input data through a sliding window method and employs a TCN model to capture temporal dependencies in MTS. After applying the TCN, the data are transformed into a latent space through linear mapping. The distribution in latent space is controlled by a modified one-class classification loss function to better handle latent representations.

Decoder: The primary function of the decoder is to reconstruct the input data from the latent representation and compute the reconstruction loss. The core of the decoder network is a self-attention mechanism, which differs from the canonical version by incorporating the GAT [47] model. This incorporation enhances the flexibility of attention score calculation. The final loss score, which combines the reconstruction loss and the one-class classification loss, effectively evaluates data abnormalities.

C. Encoder and One-Class Classification Loss

In the encoder module, two primary tasks are performed. The first task is to set the center of the data in the latent space. In this study, the method from the COUTA [53] model is adopted, where the initial encoder output point is set as the data center.

The second task involves finding a latent space representation for the decoder input, similar to standard encoder–decoder models. To better capture temporal features, the encoder of the HSC model employs both TCN and multilayer perceptron (MLP). As represented by (6), these two modules are used to derive the latent space representation

$$f_{\text{encoder}} = \left[f_{\text{TCN}}^{(K)} \circ f_{\text{TCN}}^{(K-1)} \circ \dots \circ f_{\text{TCN}}^{(1)}(\mathbf{X}) \right] \mathbf{W}_{\text{MLP1}} + \mathbf{b}_{\text{MLP1}} \quad (6)$$

where K represents the number of TCN layers used in the encoder, $\mathbf{W}_{\text{MLP1}}$ and $\mathbf{b}_{\text{MLP1}}$ are the parameters of the MLP.

To fully utilize the latent representation, HSC adopts a method similar to SVDD [12], projecting normal data into the hypersphere's interior. The loss function for this task is given by

$$\text{Loss}_{\text{orig}} = \mathbb{E}_{\mathbf{x} \sim \mathbf{X}} [\|f(\mathbf{x}) - \mathbf{c}\|^2] \quad (7)$$

where $f(\cdot)$ denotes the feature extraction process, and $\mathbf{c}$ represents the center of the hypersphere. Optimizing this loss function ensures normal data maps inside the hypersphere, while anomalies are mapped outside.

However, using (7) as the loss function can centralize normal data near the hypersphere's center, reducing its radius and causing similar representations. This diminishes unique features and compromises the effectiveness of TCN. Consequently, the downstream decoder struggles to extract reliable features for reconstruction. An optimized loss function, as shown in (8), addresses this by projecting data onto the hypersphere's surface, preserving more features and alleviating

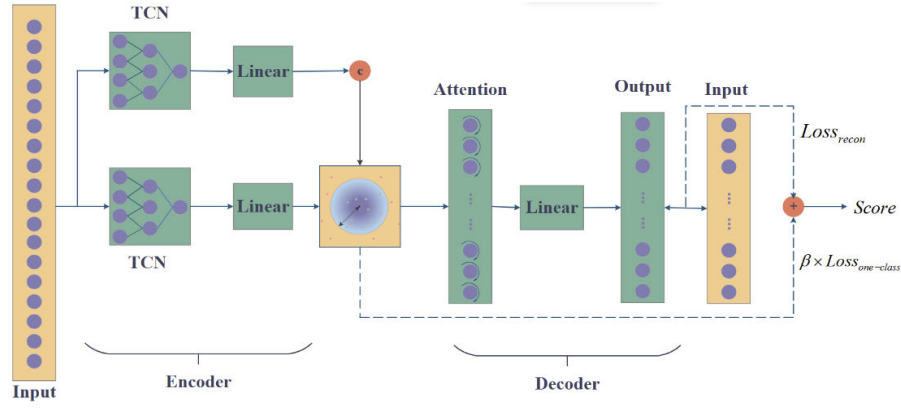


Fig. 3. Overall structure of HSC.

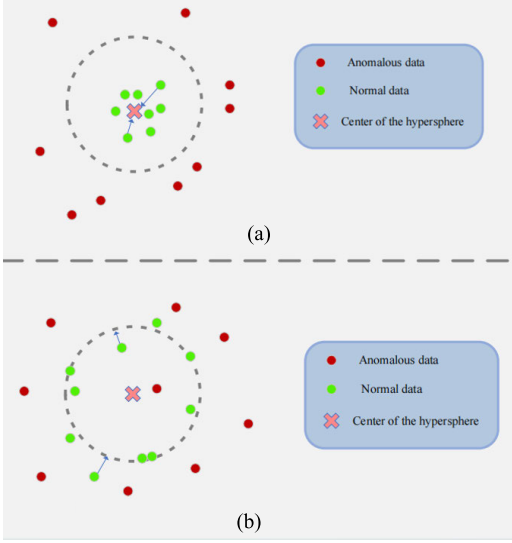


Fig. 4. Comparison of the effects of original one-class classification loss function and optimized loss function in HSC. (a) Effect of the original loss function. (b) Effect of the optimized loss function in HSC.

the issue of feature disappearance

$$\mathbf{Z} = f_{\text{encoder}}(\mathbf{X}), \quad \mathbf{Z} = (\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_N)$$

$$\text{Loss}_{\text{one-class}} = \mathbb{E}_{\mathbf{z} \sim \mathbf{Z}} (\|\mathbf{z} - \mathbf{c}\| - R)^2 \quad (8)$$

where R is the hypersphere radius, $\mathbf{z}_i \in \mathbb{R}^{W \times 1}$ represents the latent representation after the encoder module.

The effects of the original one-class loss function and the optimized loss function are illustrated in Fig. 4. Fig. 4(a) shows the effect of applying the original loss function to the latent representation, where normal data tend to centralize around the hypersphere's center during training. In contrast, Fig. 4(b) demonstrates the effect of the optimized loss function as defined by (8), where the data are projected onto the surface of the hypersphere. This approach significantly preserves more features of the data by ensuring that the unique characteristics of data point are maintained on the hypersphere's surface.

D. Decoder and Reconstruction Loss

In decoder, the primary task is to reconstruct data by using the previously obtained latent representation. The extent of reconstruction is determined by comparing how similar the

original input is to the reconstructed output. The architecture of the decoder consists of two fundamental components: a self-attention mechanism and linear mapping.

To some extent, self-attention can be viewed as a self-loop in the GAT model. The key difference between these two mechanisms lies in how they compute the attention score. As illustrated in (4), the canonical attention uses the dot product to compute the attention score. In GAT, the attention score is computed using a feedforward neural network. This method of computing the attention score is known as the “additive attention mechanism” also referred to as “Bahdanau attention” [54]. Compared with the canonical attention mechanism, this method provides greater flexibility and the ability to learn complex patterns.

In the HSC model, the attention score is calculated as follows:

$$\alpha = \sigma(\alpha_{\text{att}}^T(\mathbf{w}\mathbf{z})) \quad (9)$$

where $\mathbf{w} \in \mathbb{R}^{V \times W}$ represents a transformation matrix. The activation function $\sigma(\cdot)$ computes α , the attention score $\alpha_{\text{att}} \in \mathbb{R}^V$ represents the parameters of the feedforward network used to calculate the attention score. These parameters are updated iteratively during network training.

After obtaining the attention score, the output is calculated using a layer of linear transformation, with parameters determined by $\mathbf{w}_{\text{out}} \in \mathbb{R}^{N \times V}$. The output denoted as $\mathbf{z}' \in \mathbb{R}^N$ is calculated according to the following equation:

$$\mathbf{z}' = \sigma[\mathbf{w}_{\text{out}}(\alpha \cdot \mathbf{w}\mathbf{z})]. \quad (10)$$

HSC also incorporates a multihead attention mechanism. The multihead attention score is computed as follows:

$$\alpha_h = \frac{\exp\{\sigma[\alpha_{\text{att}_h}^T(\mathbf{w}^h\mathbf{z})]\}}{\sum_{k=1}^H \exp\{\sigma[\alpha_{\text{att}_k}^T(\mathbf{w}^k\mathbf{z})]\}} \quad (11)$$

where H represents the number of heads. The SoftMax operation is applied after calculating the attention score for each head. In the GAT model, SoftMax operation is applied across adjacent nodes. However, self-attention inherently applies a self-loop graph, where adjacent nodes essentially refer to themselves. This can result in all output attention scores converging to 1. For this situation to be avoided, the SoftMax operation is applied across each head rather than each node.

The final output of the self-attention module is computed based on the calculated scores, following the same approach as canonical attention. The output of each head is concatenated and linearly transformed using $\mathbf{W}_{\text{out}} \in \mathbb{R}^{N \times (V \cdot H)}$. After passing through an activation function $\sigma(\cdot)$ and an MLP transformation which is defined by $\mathbf{W}_{\text{MLP2}}$ and $\mathbf{b}_{\text{MLP2}}$, the final output is

$$\hat{\mathbf{x}} = \mathbf{W}_{\text{MLP2}} \left(\sigma \left[\mathbf{W}_{\text{out}} \left(\bigoplus_{h=1}^H \alpha_h \cdot \mathbf{wz} \right) \right] \right) + \mathbf{b}_{\text{MLP2}}. \quad (12)$$

The reconstructed data $\hat{\mathbf{X}} = (\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2, \hat{\mathbf{x}}_3, \dots, \hat{\mathbf{x}}_T) \in \mathbb{R}^{T \times N}$ are obtained. By comparing the reconstructed output with the input data using MSE, the reconstruction loss can be computed as follows:

$$\text{Loss}_{\text{recon}} = \frac{1}{T \times N} \sum_{t=1}^T \sum_{n=1}^N (\hat{x}_t^n - x_t^n)^2. \quad (13)$$

E. Anomaly Detection

In HSC, the final error score is calculated by combining the reconstruction loss and the one-class classification loss. A ratio hyperparameter, denoted as $\beta \in [0, 1]$, is used to calculate the final loss score, as shown in the following equation:

$$\text{Score} = \|\hat{\mathbf{x}} - \mathbf{x}\| + \beta \times (\|f_{\text{encoder}}(\mathbf{x}) - \mathbf{c}\| - R)^2. \quad (14)$$

Anomalies are determined based on the final loss score. If the score exceeds a predetermined threshold, the associated timestamp is classified as an anomaly. Most anomaly detection methods adopt a point adjustment (PA) strategy to refine their outputs and determine a suitable threshold. However, the details of the PA strategy and threshold-setting method vary depending on the application scenario. For instance, the method in [55] uses the PA%K method to adjust results for different anomaly types. In [1] and [42], the threshold is dynamically adjusted according to the anomaly scores.

This work primarily emphasizes the network framework design for efficient MTSs anomaly detection. Hence, HSC follows prior works [21], [34], [40], [53], adjusting its results and performs a grid search over potential thresholds, selecting the one yielding the best $F1$ scores.

IV. EXPERIMENTS

The effectiveness of the proposed method was validated through several experiments. First, three public datasets for MTS anomaly detection were used as standard test cases to demonstrate the method's capabilities and compare the results with other methods. Second, ablation studies assessed the impact of applying one-class constraints in the latent space and evaluated the modified attention mechanism in the decoder. Third, parameter sensitivity analysis was conducted to thoroughly explore the characteristics of HSC. Finally, a simulated penicillin dataset was used as an application test case to perform a comprehensive analysis of performance under various anomaly scenarios.

A. Datasets and Metrics

1) *Datasets*: Four datasets were employed to evaluate the effectiveness of the proposed method: Secure Water Treatment (SWaT) [56], the Mars Science Laboratory Rover (MSL)

TABLE I
DETAILS OF THE DATASETS

Dataset	SWaT	SMD	MSL	Penicillin
Training Size	496800	708377	58317	40000
Testing Size	449919	708393	73729	2400
Entity Num	1	28	27	1
Feature Num	51	38	55	16
Anomaly Rate	11.97%	4.16%	10.72%	22.1%

dataset [3], Server Machine dataset (SMD) [1], and the Penicillin Simulation dataset [57].

The first three datasets are real-world datasets commonly utilized in the field of anomaly detection for MTS. These datasets were chosen as standard test cases in this research because they cover a wide range of characteristic scenarios, ensuring the generalizability of HSC.

The penicillin simulation dataset, generated based on the simulation platform described in [57], includes 100 batches of normal data used as training data and six batches of test data. The test data were designed cover different fault causes, fault variables, and fault time zones to achieve a comprehensive validation of the proposed method. The characteristics of the four datasets are presented in Table I.

2) *Evaluation Metrics*: In the experiments, the precision (P), recall (R), $F1$, and AUC score were adopted as metrics to evaluate the performance of the proposed model. The formulas for these metrics are given by the following equations:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (15)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (16)$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (17)$$

$$\text{AUC} = \int_0^1 \frac{\text{TP}}{\text{TP} + \text{FN}} d\left(\frac{\text{FP}}{\text{FP} + \text{TN}}\right). \quad (18)$$

In the task of anomaly detection, TP refers to correctly flagged anomalies. FP denotes normal data that are incorrectly classified as anomalies, and FN represents actual anomalies that are mistakenly identified as normal data. In MTS anomaly detection, precision indicates the proportion of correctly identified anomalies among all flagged anomalies, while recall represents the proportion of detected anomalies among all actual anomalies. The $F1$ score combines precision and recall, providing a balanced measurement. AUC, which measures the area under the ROC curve, provides an aggregate performance measure across all classification thresholds.

B. Standard Test Cases

1) Benchmark Comparison:

a) *Baseline models*: The efficacy of the proposed method was compared with several state-of-the-art methods. The proposed method incorporates techniques from one-class classification, self-attention, autoencoder, and the GAT model.

TABLE II
PERFORMANCE COMPARISON OF HSC WITH OTHER BASELINE METHODS ON THREE PUBLIC DATASETS

Method	SMD				SWaT				MSL			
	F1	Precision	Recall	AUC	F1	Precision	Recall	AUC	F1	Precision	Recall	AUC
COUTA	0.926 $\pm$ 0.03	0.933 $\pm$ 0.03	0.901 $\pm$ 0.04	0.782 $\pm$ 0.03	0.827 $\pm$ 0.02	0.919 $\pm$ 0.04	0.755 $\pm$ 0.04	0.773 $\pm$ 0.02	0.913 $\pm$ 0.04	0.885 $\pm$ 0.05	0.984 $\pm$ 0.01	0.642 $\pm$ 0.03
TranAD	0.771 $\pm$ 0.02	0.995$\pm$0.04	0.652 $\pm$ 0.02	0.658 $\pm$ 0.02	0.824 $\pm$ 0.02	0.962 $\pm$ 0.03	0.719 $\pm$ 0.02	0.617 $\pm$ 0.03	0.698 $\pm$ 0.04	0.655 $\pm$ 0.05	0.945 $\pm$ 0.01	0.532 $\pm$ 0.02
GReLeN	0.887 $\pm$ 0.02	0.865 $\pm$ 0.04	0.924 $\pm$ 0.03	0.732 $\pm$ 0.03	0.819 $\pm$ 0.004	0.853 $\pm$ 0.01	<u>0.788$\pm$0.008</u>	0.642 $\pm$ 0.02	<u>0.953$\pm$0.02</u>	<u>0.937$\pm$0.04</u>	0.978 $\pm$ 0.02	0.599 $\pm$ 0.03
GDN	0.853 $\pm$ 0.04	0.889 $\pm$ 0.05	0.861 $\pm$ 0.04	0.773 $\pm$ 0.04	0.835 $\pm$ 0.02	0.967 $\pm$ 0.03	0.766 $\pm$ 0.03	<u>0.816$\pm$0.03</u>	0.951 $\pm$ 0.03	0.931 $\pm$ 0.05	<u>0.988$\pm$0.02</u>	0.603 $\pm$ 0.03
MTAD-GAT	0.908 $\pm$ 0.01	0.907 $\pm$ 0.04	0.93 $\pm$ 0.03	<u>0.824$\pm$0.03</u>	0.823 $\pm$ 0.01	0.970 $\pm$ 0.03	0.716 $\pm$ 0.03	0.801 $\pm$ 0.01	0.945 $\pm$ 0.01	0.925 $\pm$ 0.01	0.991$\pm$0.002	0.624 $\pm$ 0.02
CAE-AD	0.932 $\pm$ 0.01	0.928 $\pm$ 0.04	<u>0.945$\pm$0.03</u>	0.752 $\pm$ 0.03	0.808 $\pm$ 0.006	0.977 $\pm$ 0.02	0.689 $\pm$ 0.02	0.744 $\pm$ 0.02	0.918 $\pm$ 0.02	0.894 $\pm$ 0.02	0.984 $\pm$ 0.01	0.585 $\pm$ 0.01
Anomaly-transformer	<u>0.940$\pm$0.01</u>	0.939 $\pm$ 0.01	0.956$\pm$0.01	0.736 $\pm$ 0.01	0.821 $\pm$ 0.01	0.990$\pm$0.004	0.702 $\pm$ 0.01	0.673 $\pm$ 0.02	0.944 $\pm$ 0.04	0.925 $\pm$ 0.05	0.988 $\pm$ 0.01	0.637 $\pm$ 0.03
DeepSVDD	0.672 $\pm$ 0.03	0.728 $\pm$ 0.04	0.655 $\pm$ 0.02	0.710 $\pm$ 0.03	<u>0.845$\pm$0.01</u>	<u>0.978$\pm$0.02</u>	0.746 $\pm$ 0.02	0.623 $\pm$ 0.03	0.775 $\pm$ 0.04	0.727 $\pm$ 0.05	<u>0.988$\pm$0.02</u>	0.594 $\pm$ 0.03
HSC	0.951$\pm$0.01	<u>0.967$\pm$0.02</u>	0.941 $\pm$ 0.02	0.841$\pm$0.02	0.908$\pm$0.02	0.958 $\pm$ 0.02	0.847$\pm$0.03	0.832$\pm$0.02	0.967$\pm$0.02	0.945$\pm$0.03	0.985 $\pm$ 0.02	0.662$\pm$0.02

Therefore, eight methods were selected as benchmarks: COUTA [53], TranAD [33], GReLeN [40], GDN [23], MTAD-GAT [42], CAE-AD [34], DeepSVDD [58], and anomaly transformer [9]. Among these methods, Deep SVDD [58] and COUTA [53] utilize one-class classification for anomaly detection. TranAD [33] and anomaly transformer [9] employ transformer architectures to extract features for anomaly detection. CAE-AD [34] uses an autoencoder framework for reconstruction-based anomaly detection. Finally, GReLeN [40], GDN [23], and MTAD-GAT [42] leverage graph neural networks to achieve high-efficiency anomaly detection.

b) Results and analysis: The results presented in Table II are derived from ten independent repeated experiments to ensure the robustness and reliability of the findings. For each method, we report the average values along with the corresponding standard deviations.

The HSC model demonstrates outstanding performance across all four metrics. Specifically, it achieves the highest *F1* score on all datasets, with improvements of 1.1%, 6.3%, and 1.4% on the SMD, SWaT, and MSL datasets, respectively, compared to the second-best method, highlighting its superior anomaly detection capability. In addition to *F1* scores, the HSC model also achieves the highest AUC scores on all three datasets, with increases of 1.7%, 1.6%, and 2%. The model also shows strong competitiveness in terms of precision and recall. Moreover, across ten repeated experiments, the standard deviation is consistently below 0.04, demonstrating the robustness and stability of the HSC model. Although recall and precision did not achieve the highest scores across all datasets, they are still highly competitive.

Among the compared methods, DeepSVDD underperformed, primarily because it is not designed for MTS anomaly detection and lacks the ability to capture temporal dependencies, focusing only on inter-variable relationships. This limitation impacts its effectiveness in MTS anomaly detection. COUTA, being a purely one-class classification-based method, lacks a sufficient understanding of data distribution compared to reconstruction-based models, resulting in relatively weaker performance. CAE-AD, while using an autoencoder and contrastive learning, does not fully utilize the latent

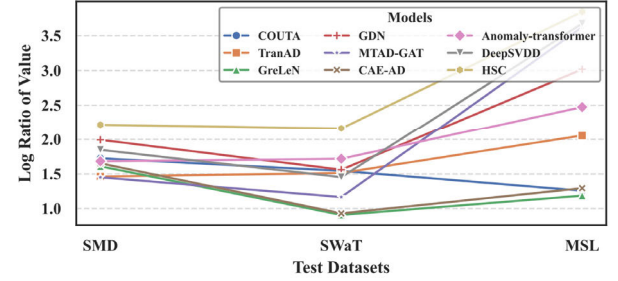


Fig. 5. Log ratio of anomaly-to-normal scores for models.

space distribution as effectively as HSC. Methods such as GDN, GReLeN, and MTAD-GAT use graph neural networks to capture inter-variable relationships. However, their focus on modeling normal data makes it harder to detect subtle anomalies. TranAD and Anomaly Transformer, based on transformer architectures, are more complex and prone to overfitting, especially when training data are limited, as seen in the SMD and MSL datasets, resulting in suboptimal performance.

The proposed HSC model is capable of capturing both temporal dependencies and intervariable relationships. By applying a HSC in the latent space, HSC effectively differentiates between normal and anomalous data, enhancing its ability to detect subtle anomalies that may be overlooked by other models. Additionally, compared to purely reconstruction-based or prediction-based models, HSC benefits from the integration of one-class classification loss and reconstruction loss, which contributes to its robustness.

c) Anomaly distinction: To validate the effectiveness of HSC in enhancing the distinction between normal and anomalous data, we first calculate the average score of normal data S_n and anomalous data S_a ; then compute the ratio of these two scores. The formula for this calculation is given by $\log(S_a/S_n)$. The results, as shown in Fig. 5, demonstrate that HSC achieves the highest distinction between normal and anomalous data across all three datasets.

d) Case study: To further demonstrate the effectiveness of latent space constraints in the HSC model, the machine-1-1

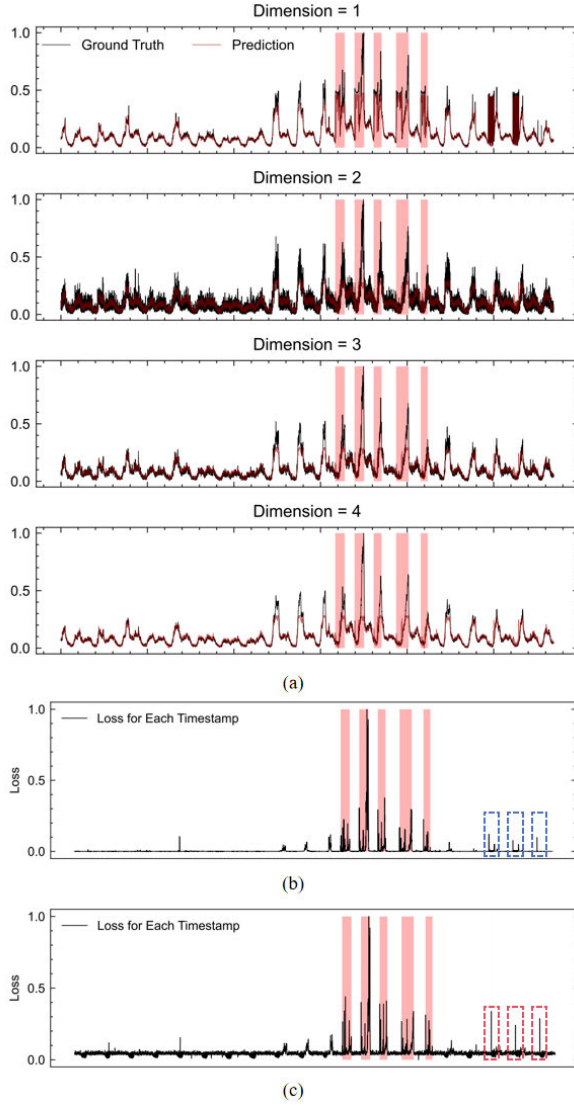


Fig. 6. Case study. (a) Visualization of four dimensions from the machine-1-1 dataset, where the black line represents the ground truth, the red line represents the model's predictions, and the five red-shaded areas indicate the time points when anomalies occurred. (b) Anomaly scores with latent space constraints applied. (c) Anomaly scores without latent space constraints.

dataset from the SMD benchmark is selected as a case study. Specific dimensions of the reconstruction results are visualized, and anomaly scores with and without latent space constraints are compared, as illustrated in Fig. 6. The model in Fig. 6(a) exhibits robust reconstruction capability for normal data. The results in Fig. 6(b) and (c) indicate that the inclusion of latent space constraints significantly improves the model's ability to distinguish between normal and anomalous data. Notably, in the absence of the HSC, as shown in Fig. 6(c), increased background noise in the anomaly scores is observed, and high anomaly scores occur after the anomaly ends. These findings highlight the importance of latent space constraints in mitigating noise interference and enhancing the accuracy of anomaly detection.

2) *Ablation Test*: The impact of constraining the distribution of representations in the latent space needs to be assessed. Comparative experiments were conducted by removing this

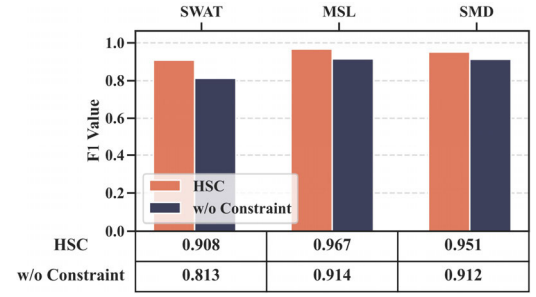


Fig. 7. Comparison of $F1$ scores between having and not having latent space restrictions across the three public datasets.

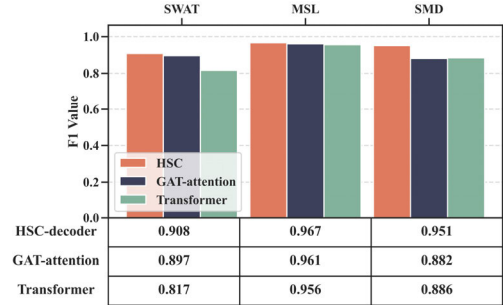


Fig. 8. Comparison of $F1$ -score between the using GAT, Transformer as the decoder of HSC, and original HSC decoder.

constraint and assessing its effects on three datasets. The results, using $F1$ as an evaluation metric, are shown in Fig. 7. After removing the constraint in the latent space, the overall average $F1$ score decreased by 6.23%. The smallest decrease was observed on the SMD dataset, with a 3.9% reduction in $F1$ score. The most significant decrease, 9.5%, was observed on the SWaT dataset. These results indicate that applying constraints within the latent space significantly enhances anomaly detection capability.

In the HSC model's decoder, self-attention was employed for feature extraction. The attention mechanisms of both GAT and the transformer were then integrated into the framework. The effectiveness of combining these two mechanisms was assessed in this study. The self-attention mechanism in the decoder was replaced by the attention mechanisms from the transformer model and GAT. As shown in Fig. 8, the HSC model achieved the best performance across the three datasets. When the decoder's self-attention module was substituted with GAT, the average $F1$ score decreased by approximately 3%. When the transformer model was used, the $F1$ score decreased by 5.6%. The most significant improvement achieved using the HSC model was on the SMD dataset. When GAT and the transformer model were used, the $F1$ scores were only 88.2% and 88.6%, respectively; by contrast, the $F1$ score of the HSC model was 95.1%. These results demonstrate that combining GAT and transformer architectures can effectively improve the model's capabilities.

3) *Hyperparameter Sensitivity Analysis*: This section analyzes the sensitivity of the proposed method to two crucial hyperparameters: the hypersphere radius R and the ratio hyperparameter β . The analysis is performed on the SWaT, MSL (D-14), and SMD (machine-1-1) datasets.

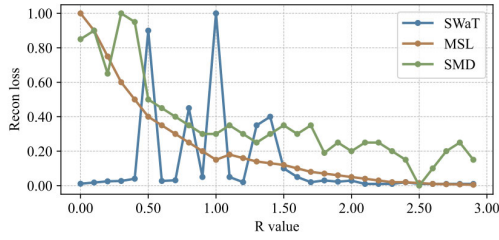


Fig. 9. Reconstruction loss on different datasets (SWaT, MSL, and SMD) with varying R values.

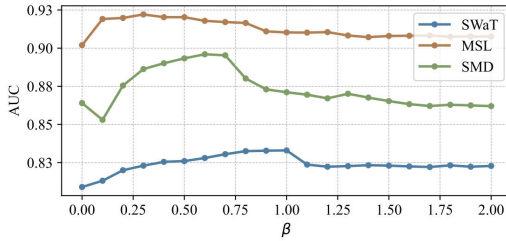


Fig. 10. AUC score on different datasets (SWaT, MSL, and SMD) with varying β values.

a) *Hypersphere radius R* : Our experiments reveal that the hypersphere radius impacts feature retention and data reconstruction. By varying R from 0 to 3 in increments of 0.1, we assessed the reconstruction loss on each dataset, as shown in Fig. 9.

- 1) *SWaT*: Small radius values cause significant fluctuations in reconstruction loss. As R increases, the loss stabilizes, indicating that a larger radius ensures consistent reconstruction performance.
- 2) *MSL*: The reconstruction loss consistently decreases with an increasing radius, suggesting that a larger R enhances the model's reconstruction ability.
- 3) *SMD*: Although some fluctuations are observed, the reconstruction loss generally decreases as the radius increases, confirming that a larger R reduces reconstruction errors.

In summary, a larger hypersphere radius R helps maintain consistent and improved reconstruction performance across different datasets.

b) *Ratio hyperparameter β* : To verify the impact of β on the results, β was varied from 0 to 2 in increments of 0.1. The AUC score under different β conditions are shown in Fig. 10.

- 1) *SWaT*: The AUC score increases as β approaches 1, after which it starts to decline. This indicates that a balanced emphasis on both losses leads to optimal anomaly detection performance.
- 2) *MSL*: The AUC score reaches its maximum at β values around 0.5. Beyond this point, the performance decreases. When β is greater than 1, the AUC score remains relatively stable.
- 3) *SMD*: The AUC score peaks at β values around 0.6, following this, the AUC score decreases similar to MSL, though the decline is more gradual.

In summary, excessively large or small β values negatively impact the results. This demonstrates the importance

TABLE III
INFORMATION ON SIX ABNORMAL SETTINGS
IN PENICILLIN SIMULATION PROCESS

No.	Abnormal Variable	Abnormal Type	Abnormal Amplitude	Introduction time
1	Aeration rate	Step	+5%	150-300h
2	Aeration rate	Ramp	+3%	200-270h
3	Agitator power	Step	+5%	260-350h
4	Agitator power	Ramp	+3%	320-400h
5	Substrate feed rate	Step	+10%	250-300h
6	Substrate feed rate	Ramp	+0.01%	260-350h

of balancing the one-class loss and reconstruction loss for the model. Setting β between 0.5 and 1 is a reasonable range.

C. Application Test Case

The capabilities of the HSC model were also evaluated in this study. In particular, the complex process of penicillin fermentation was introduced to assess the performance of the HSC model under various anomalous conditions. This process represents a typical dynamic and multistage batch production process with long operational cycles and relatively complex reactions. The penicillin fermentation simulation platform “Pensim,” developed by Professor Cinar’s research group [57]. Pensim simulates growth variations across variables under diverse production conditions, offering a standardized platform for detecting anomalies in multivariate time-series data.

In this simulation platform, the duration of the entire fermentation reaction was set to 400 h, with data sampled every hour, incorporating 16 variables. The training dataset included 100 instances of normal fermentation operations. Pensim allows for the introduction of faults in three variables: aeration rate, agitation power, and substrate feed rate. These faults are categorized into step and ramp types, and their magnitude and timing can be artificially set. During simulation, six different sets of anomalous variable data, with specific conditions for anomaly timing and types, were generated to validate the effectiveness of the HSC model, as shown in Table III.

Comparisons were also performed with COUTA [53], TranAD [33], GReLeN [40], GDN [23], MTAD-GAT [42], and CAE-AD [34], DeepSVDD [58], and anomaly transformer [9]. The $F1$ score, precision, recall, and AUC score were used as metrics. Each experiment was repeated ten times with different random seeds. The comparative results are displayed in Table IV.

Experimental results indicate that for the first two types of anomaly patterns, all models were capable of effectively locating the occurrence of faults. However, the TranAD, GReLeN, GDN, DeepSVDD, anomaly transformer, and MTAD-GAT models experienced false positives in the first four relatively straightforward test datasets, failing to achieve 100% precision. By contrast, the COUTA, CAE-AD, and HSC

TABLE IV
PERFORMANCE COMPARISON OF HSC WITH OTHER BASELINE METHODS ON SIX PENISM DATASETS

Method	Penism1				Penism2				Penism3			
	F1	Precision	Recall	AUC	F1	Precision	Recall	AUC	F1	Precision	Recall	AUC
COUTA	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.954 _{±0.03}	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.945 _{±0.03}	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.941 _{±0.02}
TranAD	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.998 _{±0.02}	<u>0.993</u> _{±0.03}	<u>0.986</u> _{±0.02}	1.00 _{±0}	<u>0.967</u> _{±0.01}	0.885 _{±0.03}	0.813 _{±0.03}	1.00 _{±0}	0.922 _{±0.02}
GReLeN	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.807 _{±0.02}	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.931 _{±0.02}	<u>0.995</u> _{±0.03}	<u>0.985</u> _{±0.03}	1.00 _{±0}	0.951 _{±0.03}
GDN	<u>0.989</u> _{±0.03}	<u>0.978</u> _{±0.02}	1.00 _{±0}	0.981 _{±0.03}	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.954 _{±0.02}	0.983 _{±0.04}	0.963 _{±0.02}	1.00 _{±0}	0.932 _{±0.03}
MTAD-GAT	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.973 _{±0.01}	0.962 _{±0.01}	0.933 _{±0.02}	1.00 _{±0}	0.937 _{±0.03}	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.953 _{±0.02}
CAE-AD	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.982 _{±0.03}	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.975 _{±0.02}	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	<u>0.966</u> _{±0.03}
DeepSVDD	0.963 _{±0.03}	0.937 _{±0.02}	1.00 _{±0}	0.732 _{±0.02}	0.968 _{±0.03}	0.933 _{±0.02}	1.00 _{±0}	0.818 _{±0.03}	0.993 _{±0.01}	0.982 _{±0.02}	1.00 _{±0}	0.827 _{±0.03}
Anomaly-transformer	0.917 _{±0.03}	0.853 _{±0.03}	1.00 _{±0}	0.683 _{±0.03}	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.916 _{±0.02}	0.962 _{±0.03}	0.983 _{±0.03}	1.00 _{±0}	0.963 _{±0.03}
HSC	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	<u>0.993</u> _{±0.02}	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.954 _{±0.02}	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.983 _{±0.02}

Method	Penism4				Penism5				Penism6			
	F1	Precision	Recall	AUC	F1	Precision	Recall	AUC	F1	Precision	Recall	AUC
COUTA	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.941 _{±0.02}	0.905 _{±0.03}	0.855 _{±0.02}	1.00 _{±0}	0.636 _{±0.02}	0.961 _{±0.02}	0.928 _{±0.03}	1.00 _{±0}	0.702 _{±0.04}
TranAD	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.995 _{±0.04}	<u>0.951</u> _{±0.02}	<u>0.907</u> _{±0.03}	1.00 _{±0}	<u>0.819</u> _{±0.04}	0.932 _{±0.03}	0.886 _{±0.02}	1.00 _{±0}	0.644 _{±0.04}
GReLeN	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.931 _{±0.04}	0.893 _{±0.02}	0.811 _{±0.05}	1.00 _{±0}	0.752 _{±0.02}	<u>0.983</u> _{±0.03}	<u>0.963</u> _{±0.03}	1.00 _{±0}	<u>0.766</u> _{±0.03}
GDN	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	<u>0.967</u> _{±0.03}	0.692 _{±0.03}	0.596 _{±0.02}	1.00 _{±0}	0.588 _{±0.03}	0.775 _{±0.03}	0.624 _{±0.03}	1.00 _{±0}	0.567 _{±0.04}
MTAD-GAT	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.913 _{±0.04}	0.932 _{±0.02}	0.893 _{±0.04}	1.00 _{±0}	0.807 _{±0.02}	0.968 _{±0.04}	0.952 _{±0.03}	1.00 _{±0}	0.699 _{±0.04}
CAE-AD	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.886 _{±0.04}	0.739 _{±0.04}	0.626 _{±0.04}	1.00 _{±0}	0.597 _{±0.04}	0.693 _{±0.02}	0.534 _{±0.02}	1.00 _{±0}	0.472 _{±0.04}
DeepSVDD	<u>0.883</u> _{±0.02}	<u>0.792</u> _{±0.03}	1.00 _{±0}	0.829 _{±0.02}	0.594 _{±0.03}	0.464 _{±0.03}	1.00 _{±0}	0.456 _{±0.03}	0.714 _{±0.03}	0.522 _{±0.03}	1.00 _{±0}	0.522 _{±0.04}
Anomaly-transformer	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.933 _{±0.04}	0.495 _{±0.02}	0.368 _{±0.01}	1.00 _{±0}	0.422 _{±0.03}	0.732 _{±0.03}	0.573 _{±0.02}	1.00 _{±0}	0.458 _{±0.03}
HSC	1.00 _{±0}	1.00 _{±0}	1.00 _{±0}	0.944 _{±0.02}	0.989 _{±0.03}	0.981 _{±0.04}	1.00 _{±0}	0.871 _{±0.03}	0.985 _{±0.04}	0.971 _{±0.02}	1.00 _{±0}	0.853 _{±0.03}

models performed flawlessly on the first four datasets, captured anomalies with precision, and achieved an $F1$ score of 100%.

In the last two datasets, where anomalies were subtler and posed greater detection challenges. The HSC model demonstrated its superior anomaly detection capability. Specifically, HSC achieved the highest $F1$ scores on both the Penism5 and Penism6 datasets, with scores of 98.9% and 98.5%, respectively. The AUC scores for HSC were also the highest: 87.1% on Penism5 and 85.3% on Penism6, representing improvements of 5.2% and 8.7% over the next best methods, respectively.

a) *Anomaly distinction*: In Penism dataset, to demonstrate the distinction between normal and anomalous data for the HSC model, the same calculation was performed for different models on various datasets. The anomaly distinction score, $\log(S_a/S_n)$, is presented in Fig. 11. On all six test datasets, the HSC model consistently achieved the highest scores. On the first four datasets, all models could effectively distinguish between normal and anomalous patterns. However, the models' capabilities diverged on the Penism5 and Penism6 datasets. In particular, GDN, CAE-AD, GReLeN, DeepSVDD, and anomaly transformer, exhibited negative scores which

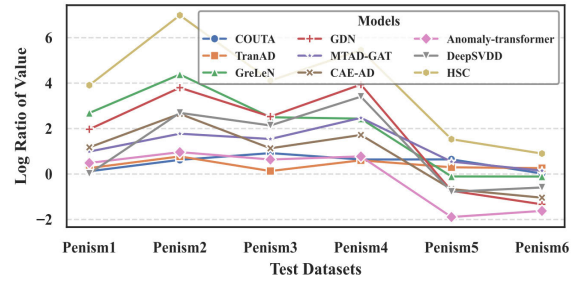


Fig. 11. Log ratio of anomaly to normal data scores across models on six Penism test datasets.

indicate an inability to effectively discriminate between normal and anomalous data.

V. DISCUSSION

In this section, the discussion focuses on the potential challenges the HSC model may encounter in real-world industrial environments, particularly regarding its scalability and robustness to noise. In addition, the section provides an analysis of potential future research directions for the HSC model.

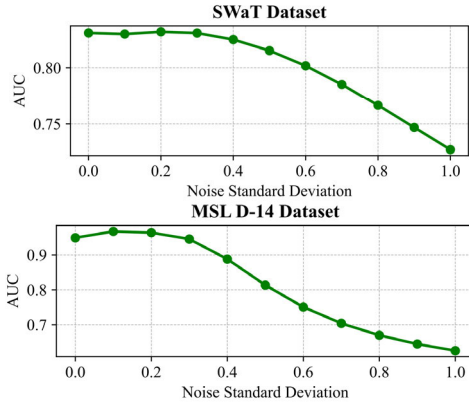


Fig. 12. AUC variation of the SWaT dataset and MSL D-14 dataset under different noise levels.

A. Scalability

The primary computational bottleneck in the HSC model stems from two factors: the 1-D convolution operation in the TCN and the self-loop attention mechanism.

The time complexity of the TCN's 1-D convolution is $O(W \cdot k \cdot N)$, where W represents the sliding window length, k is the kernel size, and N is the feature dimension. Given its linear complexity, the model exhibits good scalability for handling long sequence data.

For the self-loop attention mechanism, the time complexity of embedding transformation is $O(W \cdot N^2)$. The complexity of computing attention scores is $O(W \cdot N)$, while applying attention weights has a complexity of $O(W \cdot N^2)$. Given that the window size can be adjusted, the complexity is primarily constrained by the number of variables N .

Overall, the computational bottleneck in the HSC model is largely determined by the number of variables in the data. Nevertheless, it is particularly well suited for capturing global relationships within the data, which are critical for MTS anomaly detection tasks.

B. Robustness to Noise

To assess the robustness of the HSC model in industrial scenarios, robustness experiments were performed on the SWaT and MSL D-14 datasets. Gaussian noise with varying intensities was introduced to both datasets, where the noise intensity was controlled by the standard deviation (std) of the Gaussian distribution. The std ranged from 0 to 1, increasing in steps of 0.1. Each experiment was repeated five times, and the average AUC scores were calculated for analysis. The results are shown in Fig. 12.

As depicted in Fig. 12, when the noise level is low (std ≤ 0.1), the model's performance remains stable, with AUC scores showing minimal variation, demonstrating strong robustness. In moderate noise scenarios ($0.1 < \text{std} \leq 0.4$), slight fluctuations are observed; however, the overall performance remains satisfactory. However, under high noise conditions (std > 0.4), the AUC scores drop significantly, indicating that the model is sensitive to high levels of noise.

In conclusion, the HSC model demonstrates a certain level of robustness to noise; however, its performance

degrades in environments characterized by high levels of noise interference.

C. Future Research Directions

The HSC model has demonstrated strong capabilities in detecting MTS anomalies. However, to enhance its effectiveness in real-world production settings and to reduce losses, there remains room for improvement.

First, although the HSC model exhibits strong performance in various experiments, its reconstruction effectiveness is sensitive to the radius of the hypersphere. Different datasets require distinct hypersphere radii, and currently, there is no effective method to determine the optimal radius. This necessitates manual tuning for each dataset based on specific application scenarios. We infer that this sensitivity may stem because imposing constraints in the latent space inevitably leads to the loss of certain important features. Therefore, future research could focus on exploring ways to limit the distribution of data in the latent space while minimizing feature loss, in order to reduce the model's sensitivity to hyperparameters.

Second, current MTS anomaly detection primarily focuses on identifying the moment when an anomaly occurs, providing early warnings. However, to more effectively reduce losses, early warnings alone are insufficient; root cause analysis is also required. Through root cause analysis, the specific cause and location of the anomaly can be quickly identified, enabling timely intervention and minimizing potential losses. Therefore, future research could further explore how to integrate anomaly detection with root cause analysis, enhancing the practical utility and efficiency of handling anomalies.

VI. CONCLUSION

This article proposes a new network architecture, termed HSC, designed to address the challenge of anomaly detection in MTS. First, HSC combines the one-class classification with reconstruction-based deep learning method to accentuate the distinctions between normal and anomalous data. Second, HSC leverages a self-attention mechanism, integrating the Transformer and GAT, to enhance input data reconstruction. In the end, by combining the one-class classification loss and reconstruction loss, HSC computes the final loss score, which serves as a basis for anomaly detection.

Extensive experiments conducted on multiple datasets demonstrate that the HSC model offers significant advantages over baseline models, particularly in its ability to enhance the distinction between normal and anomalous data, thus improving anomaly detection performance. The effectiveness of the model's components has been validated through ablation studies, which underscore the contributions of each module to the overall performance. Furthermore, noise experiments demonstrate the model's robustness, indicating its capability to perform reliably even in complex and noisy environments.

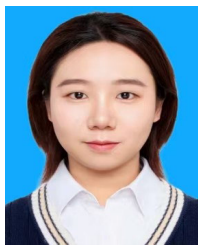
Future research may focus on developing more efficient approaches for constraining data distribution within the latent space, with an emphasis on minimizing feature loss and reducing the model's sensitivity to hyperparameters. Additionally, incorporating root cause analysis into the framework could

enable the rapid identification of anomaly causes and locations, further enhancing the model's ability to provide early warnings and support timely intervention.

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